

Eigenvalues and Eigenvectors

Field F : Real $\mathbb{R}$ or Complex $\mathbb{C}$.

Invariant Subspaces

← A linear map $T: V \rightarrow V$ is called an operator $T \in L(V)$.

[Suppose $V = V_1 \oplus \dots \oplus V_k$, $\vec{x}_i \in V_i$. $T\vec{x}_i \in V$
but there's no guarantee that $T\vec{x}_i \in V_i$.
 $T|_{V_i}$ denotes action of T on $V_i \subset V$.

← Invariant subspace:

Suppose $T \in L(V)$. A subspace $\mathcal{U}$ of V is called invariant under T if
 $\forall \vec{u} \in \mathcal{U}, T\vec{u} \in \mathcal{U}$.

[$\mathcal{U}$ is invariant under T if $T|_{\mathcal{U}} \in L(\mathcal{U})$.
If $T \in L(V)$, following subspaces of V are all invariant under T :

(i) $\{0\}$

(ii) V

(iii) $\text{null}(T) := \text{kernel}(T)$

(iv) $\text{Image}(T) := \text{range}(T)$

Q. Must $T \in L(V)$ have any invariant subspaces other than $\{0\}$ and V ?

Yes, if $\dim(V) < \infty$ and $\dim V > 1$ (for $F = \mathbb{C}$)
or $\dim V > 2$ (for $F = \mathbb{R}$).

✓ Invariant subspaces of dim 1.

For any $\vec{v} \in V$, $\vec{v} \neq 0$,

$$U := \{\alpha \vec{v} : \alpha \in F\} = \text{span}(\vec{v})$$

U is 1-dim subspace of V . Every 1-dim subspace of V is of this form for apt. $\vec{v}$.

If $U = \text{span}(\vec{v})$ is invariant under T , i.e., $T|_U \in L(U)$, then $T\vec{v} \in U$,

$$\text{i.e., } \exists \lambda \in F \text{ s.t. } T\vec{v} = \lambda \vec{v}.$$

Conversely, if $T\vec{v} = \lambda \vec{v}$ for some $\lambda \in F$, then $U = \text{span}(\vec{v})$ is invariant under T .

✓ $T \in L(V)$. $\lambda \in F$ is called an eigenvalue of T if $\exists \vec{v} \in V$ s.t. $T\vec{v} = \lambda \vec{v}$, $\vec{v} \neq \vec{0}$.
 $\vec{v}$ is called an eigenvector of T corresponding to $\lambda \in F$ when $T\vec{v} = \lambda \vec{v}$ for $\vec{v} \neq \vec{0}$.

T has a 1-D subspace invariant under $T \in L(V)$ iff T has an eigenvalue.

Example: $T \in L(F^3)$

$$T(x, y, z) = (7x + 3z, 3x + 6y + 9z, -6y)$$

$$\neq (x, y, z) \in F^3.$$

$$T(3, 1, -1) = (18, 6, -6) = \underbrace{6(3, 1, -1)}_{\text{eigenvector}} \text{ eigenvector}$$

↗ eigenvalue

Thm. (Equivalent conditions to be an eigenvalue)
 Suppose $\dim V < \infty$, $T \in L(V)$, $\lambda \in F$. Then the following are equivalent:

- (a) λ is an eigenvalue of T .
- (b) $T - \lambda I$ is not one-to-one (injective).
- (c) $T - \lambda I$ is not surjective (onto).
- (d) $T - \lambda I$ is not invertible.

Because $T\vec{v} = \lambda\vec{v}$ iff $(T - \lambda I)\vec{v} = \vec{0}$, $\vec{v} \in V$ with $\vec{v} \neq \vec{0}$ is an eigenvector of T corresponding to $\lambda \in F$ iff $\vec{v} \in \text{null}(T - \lambda I)$.

Examples: $T \in L(F^2)$, $T(\omega, \zeta) = (-\zeta, \omega)$.

(a) $F = \mathbb{R}$. $T(r\cos\theta, r\sin\theta) = (-r\sin\theta, r\cos\theta)$
 $\hookrightarrow$ counter-clockwise rotation by 90° .

$T(1,0) = (0,1)$, $T(0,1) = (-1,0)$
 T has no eigenvalue as there's no eigenvector.

(b) $F = \mathbb{C}$. $T(\omega, \zeta) = \lambda(\omega, \zeta)$, say $(\omega, \zeta) \neq (0,0)$.
 or, $(-\zeta, \omega) = \lambda(\omega, \zeta)$

$$\Rightarrow \zeta = -\lambda\omega, \quad \omega = \lambda\zeta$$

Substituting $\omega = \lambda\zeta$ to $\zeta = -\lambda\omega$,

$$\zeta = -\lambda^2 \zeta$$

$$\text{or, } \zeta(1 + \lambda^2) = 0 \Rightarrow \lambda = \pm i$$

$$\left. \begin{aligned} T(i\zeta, \zeta) &= (-\zeta, i\zeta) = i(i\zeta, \zeta) \\ T(-i\zeta, \zeta) &= (-\zeta, -i\zeta) = -i(-i\zeta, \zeta) \end{aligned} \right\} \begin{array}{l} \text{eigenvalues} \\ \text{eigenvectors} \end{array}$$

linearly independent eigenvectors

Thm. Suppose $T \in L(V)$. Then every list of eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.

Proof: Let $\vec{v}_1, \dots, \vec{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of T .

Consider $\{\vec{v}_1, \dots, \vec{v}_r\}$ is linearly dependent.
Let k be the smallest integer (+ve) s.t.
 $\vec{v}_k \in \text{Span}\{\vec{v}_1, \dots, \vec{v}_{k-1}\}$.

$$\exists a_1, \dots, a_{k-1} \in F \text{ s.t.} \\ \vec{v}_k = \sum_{i=1}^{k-1} a_i \vec{v}_i \quad \text{--- (I)} \Rightarrow \lambda_k \vec{v}_k = \sum_{i=1}^{k-1} a_i \lambda_k \vec{v}_i \quad \text{--- (II)}$$

Applying T on both sides,

$$\lambda_k \vec{v}_k = \sum_{i=1}^{k-1} a_i \lambda_i \vec{v}_i \quad \text{--- (III)}$$

$$\text{(I)} - \text{(III)} \Rightarrow \vec{0} = \sum_{i=1}^{k-1} a_i (\lambda_i - \lambda_k) \vec{v}_i \quad \text{--- (IV)}$$

~~That is, $\exists a_i \in F$~~

k is smallest +ve integer satisfying
 $\vec{v}_k \in \text{Span}\{\vec{v}_1, \dots, \vec{v}_{k-1}\}$,

implying $\{\vec{v}_1, \dots, \vec{v}_{k-1}\}$ is linearly independent.

(IV) implies all a_i 's are 0 ($\because \lambda_i$'s are distinct).

This means $\vec{v}_k$ is $\vec{0}$ (from (I)), but $\vec{v}_k \neq \vec{0}$.

$\therefore \{\vec{v}_1, \dots, \vec{v}_k\}$ can't be linearly dependent.

Corollary: No. of distinct eigenvalues of $T \leq \dim(V)$.

Restriction & Quotient operators:

If $T \in L(V)$ & U is a subspace of V invariant under T , then U determines two other operators $T|_U \in L(U)$ & $T/U \in L(V/U)$.

① $T|_U(\vec{u}) = T\vec{u} \quad \forall \vec{u} \in U$. ~~$T|_U$~~ $T|_U$: restriction op.

② $T/U \in L(V/U)$, $\rightarrow$ Quotient operator:

$$T/U(\vec{v}+U) = T\vec{v}+U \quad \forall \vec{v} \in V$$

If $\vec{v}+U = \vec{w}+U$, then $T\vec{v}+U \stackrel{?}{=} T\vec{w}+U$.
suppose $\vec{v}+U = \vec{w}+U \Rightarrow (\vec{v}-\vec{w})+U = U$
 $\Rightarrow \vec{v}-\vec{w} \in U$.

$$\Rightarrow T(\vec{v}-\vec{w}) \in U$$

$$\Rightarrow T\vec{v}+U = T\vec{w}+U.$$